

No. of Functional Features Included in the Solution

Date	15 March 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration & Login	Allows applicants to securely register (email/socials) and log into the system to apply for loans.	Web App / Auth Service	Done	0.5
2	Exploratory Data Analysis	Generates univariate/bivariate analysis and correlation heatmaps to understand feature relationships.	analysis.py	Done	0.5
3	Automated Data Preprocessing	Imputes missing data, encodes categorical strings, applies StandardScaler, and balances classes using SMOTE.	preprocessing.py	Done	1.0

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4	Automated Model Selection	Trains multiple algorithms (Decision Tree, RF, KNN, XGBoost) and selects the one with the highest accuracy.	train_model.py	Done	1.0
5	Model Serialization	Decouples ML from the web server by saving the best model and scaler into lightweight .pkl files.	train_model.py / Storage	Done	0.5
6	Dynamic Web Interface	Renders responsive frontend pages (home.html, predict.html, submit.html) for user input.	app.py / Frontend	Done	0.5
7	Real-Time Prediction Inference	Captures POST data via /submit, scales it using the loaded model, and returns an Approved/Rejected decision.	app.py / Backend	Done	0.5
8	Database & Records Management	Stores CSV datasets, serialized models, and maps applicant profile data relationally.	Data Layer	Done	0.5

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	8
Core / Must-Have Features	6 (Preprocessing, Training, Serialization, UI, Inference, DB)

Feature Summary:

Additional / Nice-to-Have Features	2 (EDA Visualizations, Social Login options)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Dynamic Web Interface (home.html, predict.html, submit.html)
2	Backend / Logic	4	Automated Data Preprocessing, Automated Model Selection, RealTime Prediction Inference
3	Database / Storage	2	Database & Records Management, Model Serialization (.pkl exports)
4	API / Integration	1	Connecting the Flask /submit route to the serialized ML pipeline
5	Security / Authentication	1	User Registration & Login (Email, Password, Auth verification)